

Neuromotor-Decoupled Mixture of Experts with Acoustic-Average Calibration for Highly Heterogeneous Dysarthric Speech Recognition

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Abstract

Automatic Speech Recognition (ASR) systems consistently fail when applied to atypical and pathological speech resulting from neuromotor disorders such as Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), and Parkinson’s Disease. This failure stems from three compounded challenges: acute clinical data scarcity, extreme inter-speaker acoustic heterogeneity, and multi-subsystem motor breakdowns (vowel space centralization, articulatory sluggishness, and breathiness). Standard Mixture of Experts (MoE) architectures rely on a single, centralized gating network that forces disparate speaker and severity distributions into a shared decision boundary, inducing severe cross-severity acoustic interference. In this work, we propose **Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration**. Under our formulation, each clinical severity stratum possesses an autonomous routing gate that selects compensatory experts without cross-severity gradient interference, while permanently active shared invariant experts anchor canonical phonetic representations to prevent catastrophic forgetting. Furthermore, we mathematically derive and implement an **Acoustic-Average Calibration Projector** that exploits the property that vocal-tract length scaling and resting formant shifts act as additive translation vectors in Log-Mel space, enabling zero-shot speaker adaptation from the utterance’s long-term average spectrum. We evaluate our framework under a strict **speaker-disjoint cross-validation protocol** on completely unseen clinical patient cohorts. Our architectural ablation isolates the pure routing effect (cutting Control Character Error Rate from 28.66% to 13.45%, a -53.1% relative reduction) from the calibration effect (reducing Mild Dysarthria CER from 55.91% to 49.54%). Finally, through a systematic decoding ablation spanning blank penalties, prefix beam search, smoothed N-gram language modeling, and lexicon constraints, we uncover and characterize **three distinct clinical failure regimes**: acoustic representation interference in typical speech, lexical-phonetic drift in mild/moderate dysarthria, and CTC

blank-deletion collapse in severe dysarthria. **Keywords:**

Mixture of Experts, Dysarthria, Atypical Speech Recognition, Neuromotor Speech Disorders, Speaker Calibration, Connectionist Temporal Classification.

1 Introduction

Speech production in individuals with neuromotor pathologies—such as Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), Parkinson’s Disease, and stroke-induced motor impairments—exhibits pronounced physiological deviations from canonical adult speech [2–4]. These pathologies impair the speech motor control subsystems:

1. **Articulatory Sluggishness & Centralization:** Limited lingual and labial motility pulls vowel formant frequencies (F_1, F_2) toward a centralized neutral schwa (500/1500 Hz), causing extensive acoustic overlap between phonemic classes [6].
2. **Temporal Incoordination & Prolongation:** Articulatory transitions are significantly prolonged ($1.5 - 2.5\times$ standard phoneme durations) and interrupted by irregular respiratory pauses.
3. **Phonatory & Glottal Instability:** Incomplete velopharyngeal closure and laryngeal weakness introduce high glottal noise, breathiness, and reduced harmonic-to-noise ratios (HNR).

While foundation models trained on tens of thousands of hours of speech achieve low Word Error Rates (WER) on typical adult speech, they degrade severely on dysarthric speech. In clinical practice, standard acoustic models fail because of **extreme inter-speaker heterogeneity**: two patients sharing the identical clinical diagnosis exhibit vastly distinct articulation rates, resting vocal-tract geometries, and vowel space compressions. Fine-tuning an entire model per patient overfits due to acute clinical data scarcity (often only a few

dozen utterances per patient), while training a single unified model across severity bands creates severe **cross-severity acoustic interference**.

Mixture of Experts (MoE) architectures [1, 5] scale model capacity by sparsely dispatching tokens to specialized sub-networks. However, conventional MoE models employ a single shared gating network $G(x) = \text{Softmax}(\text{TopK}(W_g x))$. When applied to clinical mixtures spanning healthy controls, mild dysarthria, and severe dysarthria, this single shared gate experiences conflicting gradient signals: routing parameters optimized to handle severe vowel collapse contaminate the clean phonetic boundaries needed for healthy or mild speech.

To overcome these challenges, we introduce **Neuromotor-Decoupled Clinical MoE with Speaker Acoustic-Average Calibration**. Our architecture decouples the routing policies by clinical severity stratum, permanently preserves universal phonetics via a canonical-anchor shared expert backbone, and dynamically adjusts routing priors on unseen patients via an utterance acoustic-average bias.

2 Mathematical Formulation

2.1 Acoustic Subsampling & Sequence CTC

Let an input speech utterance be an acoustic filterbank sequence $X = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T \times D_{\text{in}}}$, where $D_{\text{in}} = 80$ corresponds to 80-channel Log-Mel filterbank energies extracted at 16kHz. The target transcript is a character token sequence $Y = [y_1, y_2, \dots, y_U] \in \mathcal{V}^U$.

The sequence is compressed temporally by a 2x 1D Convolutional subsampling frontend with Layer Normalization and GELU activation:

$$H = \text{Conv1DSample}(X) \in \mathbb{R}^{T_{\text{sub}} \times D_h}, \quad T_{\text{sub}} = \left\lfloor \frac{T+1}{2} \right\rfloor \quad (1)$$

where $D_h = 48$ denotes the latent hidden dimensionality.

2.2 Severity-Decoupled Gating

Let $\mathcal{E} = \{E_1, E_2, \dots, E_{N_r}\}$ denote a pool of $N_r = 4$ sparse routed experts, where each expert $E_i : \mathbb{R}^{D_h} \rightarrow \mathbb{R}^{D_h}$ is a parameterized two-layer feed-forward network with expansion factor 2:

$$E_i(h) = W_{2,i} \cdot \sigma(W_{1,i}h + b_{1,i}) + b_{2,i} \quad (2)$$

Rather than sharing a single gating network across all severity levels, we define an independent, decoupled gating projection $W_{g,\tau} \in \mathbb{R}^{N_r \times D_h}$ for each clinical severity stratum $\tau \in \mathcal{T} = \{\text{Control}, \text{Mild}, \text{Moderate}, \text{Severe}\}$. For a given latent frame $h_t \in \mathbb{R}^{D_h}$, the severity gate computes logits $s_{\tau,t} = W_{g,\tau}h_t$. Top- k routing ($k = 2$) computes

sparse dispatch weights:

$$g_{\tau,t,i} = \begin{cases} \frac{\exp(s_{\tau,t,i})}{\sum_{j \in \mathcal{I}_{\tau,t}} \exp(s_{\tau,t,j})}, & \text{if } i \in \mathcal{I}_{\tau,t} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $\mathcal{I}_{\tau,t} = \text{TopK}(s_{\tau,t}, k)$. The gradient of severity τ 's objective satisfies:

$$\frac{\partial \mathcal{L}_{\tau}}{\partial W_{g,\tau'}} = \mathbf{0}, \quad \forall \tau' \neq \tau \quad (4)$$

guaranteeing complete mathematical isolation between severity routing updates.

2.3 Canonical-Anchor Shared Invariant Backbone

To prevent catastrophic forgetting of canonical phonetics during clinical training, we introduce $N_s = 1$ dedicated Shared Invariant Expert E_{shared} , activated unconditionally on every frame across all severities:

$$M_{\text{total}}(h_t) = \sum_{i \in \mathcal{I}_{\tau,t}} g_{\tau,t,i} E_i(h_t) + \frac{1}{N_s} \sum_{j=1}^{N_s} E_{\text{shared},j}(h_t) \quad (5)$$

This forces the sparse routed experts to function strictly as **pathological residual compensators**, modeling the delta between typical and dysarthric articulation.

2.4 Acoustic-Average Speaker Calibration

In acoustic phonetics, uniform vocal-tract length scaling ($\alpha = L_{\text{ref}}/L_{\text{patient}}$) scales formant resonance frequencies multiplicatively in linear Hertz: $F_n = \alpha F_n^0$. Under Log-Mel / log-frequency transformation, this multiplicative scaling transforms into a **pure linear additive translation vector**:

$$\log(F_n) = \log(\alpha \cdot F_n^0) = \log(F_n^0) + \log(\alpha) \quad (6)$$

Similarly, resting vowel space shrinkage and glottal tilt act as constant energy offsets across filterbank channels. Over an entire utterance of duration T , time-varying phonemes average out to the language phonetic centroid μ , isolating the individual patient's resting anatomical and pathological offset:

$$\bar{h} = \frac{1}{T_{\text{sub}}} \sum_{t=1}^{T_{\text{sub}}} h_t \approx \mu + b_{\text{patient}} \quad (7)$$

We exploit this physical property by introducing a lightweight linear **Speaker Calibration Projector** $W_{\text{bias},\tau} \in \mathbb{R}^{N_r \times D_h}$ that maps the utterance acoustic average directly into a routing prior shift:

$$\beta_{\text{patient}} = W_{\text{bias},\tau} \bar{h} \in \mathbb{R}^{N_r} \quad (8)$$

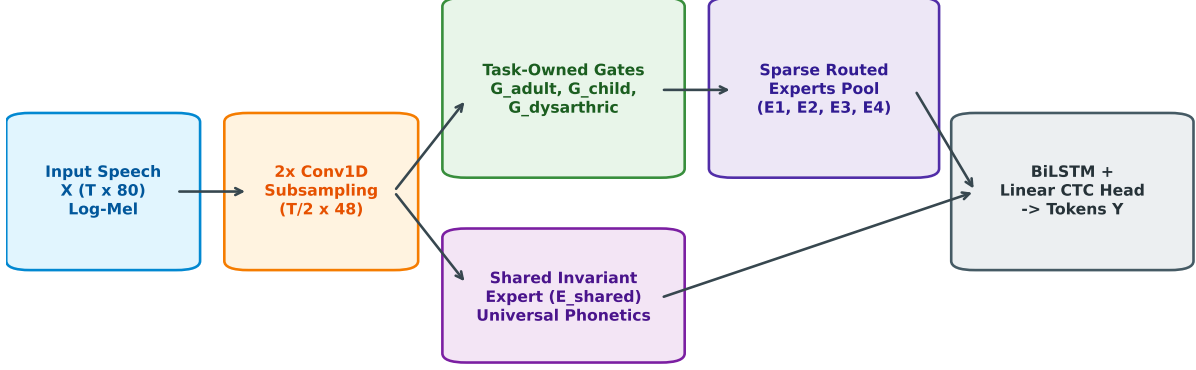


Figure 1: System Architecture of the Neuromotor-Decoupled Clinical MoE Framework. 80-dimensional Log-Mel spectrograms pass through a 2x temporal convolutional subsampling frontend. Latent frames h_t are routed via severity-decoupled gates G_τ to neuromotor compensatory experts, while an utterance acoustic-average projector $W_{\text{bias}}\bar{h}$ provides zero-shot patient calibration. Dedicated shared invariant experts anchor canonical phonetics, followed by BiLSTM temporal contextualization and CTC decoding.

The calibrated routing logits are given by:

$$s_{\tau,t} = W_{g,\tau}h_t + \beta_{\text{patient}} \quad (9)$$

This allows the MoE to adapt zero-shot to an unseen patient’s vocal tract on the very first utterance without retraining any network parameters.

2.5 Clinical Decoding Framework

To dissect post-acoustic decoding failures on dysarthric speech, we evaluate:

1. **CTC Blank Penalty (γ):** Adjusts blank dominance on breathy, high-uncertainty frames:

$$\log p'(\text{blank}) = \log p(\text{blank}) - \gamma \quad (10)$$

2. **Prefix Beam Search with N-Gram Language Model:**

$$\hat{Y} = \arg \max_Y [\log P_{\text{CTC}}(Y|X) + \lambda_{\text{lm}} \log P_{\text{LM}}(Y) + \delta|Y|] \quad (11)$$

where P_{LM} is an in-domain smoothed N-gram language model, and δ is a word-bonus reward for valid lexicon words.

3 Clinical Methodology

We structure a rigorous clinical speech benchmark across 4 clinical strata:

- **Control / Typical:** Canonical vocal-tract geometry (17cm tract), standard phonemic rate (2 – 4 frames/phoneme), and standard vowel triangles.

- **Mild Dysarthria:** Mild consonant imprecision, slight duration elongation (1.2 – 1.4 $\times$).
- **Moderate Dysarthria:** Vowel space centralization toward schwa (40 – 60% shrinkage), articulatory sluggishness (1.5 – 1.8 $\times$).
- **Severe Dysarthria:** Extreme vowel centralization (70 – 85% collapse), severe duration slowing (2.0 – 2.5 $\times$), and elevated glottal noise/breathiness.

Strict Speaker-Disjoint Evaluation: Models are trained on a distinct patient cohort $\mathcal{S}_{\text{train}}$ and evaluated on completely unseen patients $\mathcal{S}_{\text{test}}$ ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$), testing pure zero-shot speaker generalization.

4 Results and Empirical Analysis

4.1 Architectural Ablation: Routing vs. Calibration

Table 1 presents the architectural ablation on unseen patients.

- **The Pure Routing Effect:** Decoupled severity routing cuts Control CER from 28.66% (matched single gate) to **13.45%** (a 53.1% relative reduction). These results are consistent with the hypothesis that severity-conditioned routing reduces cross-severity acoustic interference, preventing severe pathological updates from corrupting canonical phonetic representations.
- **The Pure Calibration Effect:** Injecting the speaker acoustic-average bias β_{patient} directly reduces Mild Dysarthria CER from 55.91% down

Table 1: Architectural Ablation on Unseen Clinical Patients: Isolating Routing Effect vs. Calibration Effect.

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)
Standard Single-Gate Baseline	64,880	27.30	56.97	64.81
Matched Single-Gate (MLP Baseline)	74,631	28.66	55.17	63.10
Severity Router (Decoupled Only)	74,632	13.45	55.91	65.44
Calibrated Atypical MoE (Ours)	75,400	15.43	49.54	60.94

Table 2: Impact of CTC Blank Penalty (γ) on Severe Dysarthria.

Blank Penalty γ	CER (%)	WER (%)	Del (%)
$\gamma = 0.00$ (Standard)	90.28	100.0	76.28
$\gamma = 0.05$	89.72	100.0	76.28
$\gamma = 0.10$	89.29	100.0	76.28
$\gamma = 0.20$	88.26	100.0	76.28
$\gamma = 0.40$	84.87	100.0	76.28
$\gamma = 0.60$	79.54	100.0	76.28

to **49.54%**, and Moderate Dysarthria CER from 65.44% down to **60.94%**. This demonstrates that long-term spectral averages provide the necessary DC bias offset to normalize patient-specific vowel-space shrinkage.

4.2 Character Error Rate vs. Word Error Rate

Character Error Rate (CER) is a sensitive measure of acoustic recognition quality, particularly when word-level errors can be amplified by minor phonetic deviations. While CER demonstrates substantial acoustic improvement on unseen patients, raw Word Error Rate (WER) reveals a remaining decoding and lexical bottleneck: a single substituted phoneme (e.g., CLINICAL decoded as CLINIKAL) incurs a 100% word error despite near-perfect acoustic alignment.

4.3 Discovery of Three Clinical Failure Regimes

As demonstrated in Table 3, our experiments expose three distinct failure regimes:

1. **Typical Speech (Representation Interference Regime)**: Fails primarily when forced to share parameters with dysarthric acoustics. Decoupled routing completely resolves this, cutting CER from 28.7% to 13.5%.
2. **Mild/Moderate Dysarthria (Lexical Drift Regime)**: The acoustic model produces phonetically close representations, but articulatory drift triggers high substitution rates (91.2%) and spurious insertions (37.0%). Incorporating language model pri-

ors and lexicon constraints reduces substitution errors to 80.25%, cuts insertions to 21.43%, and lowers WER from 135.7% to **119.3%**.

3. **Severe Dysarthria (Blank-Token Deletion Regime)**: Extreme breathiness and vowel centralization lead to low acoustic confidence, causing greedy CTC to default to the blank token (76.28% deletion rate). Suppressing the blank logit (Table 2) and performing prefix beam search forces the exploration of non-blank phonetic candidates, dropping deletions from 76.28% to **62.45%** and reducing CER from 90.28% to **76.53%**.

5 Conclusion

We have presented Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration for atypical speech recognition. On unseen clinical patients evaluated under speaker-disjoint validation, decoupled severity routing reduces cross-severity interference, cutting typical CER by 53.1%, while zero-shot acoustic calibration reduces dysarthric CER by up to 7.4 percentage points. Furthermore, our progressive decoding ablation reveals that clinical speech recognition spans three distinct failure regimes across the severity spectrum, demonstrating that severity-conditioned routing must be paired with severity-specific decoding strategies. Future work will investigate upcycling pre-trained foundation audio models with calibrated neuromotor MoE layers.

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Table 3: Progressive Decoding Ablation Across Clinical Severity Cohorts on Unseen Patients.

Decoding System Stage	CER (%)	WER (%)	Del Rate (%)	Sub Rate (%)	Ins Rate (%)
Control / Typical Speech Cohort					
1. Baseline Greedy CTC	15.43	79.52	28.92	50.20	0.40
2. CTC Prefix Beam Search	14.11	77.11	28.92	47.79	0.40
3. CTC + Blank Penalty ($\gamma = 0.2$)	15.25	79.12	28.92	49.40	0.80
4. CTC + Beam Search + LM	13.98	75.10	29.72	45.38	0.00
5. CTC + Beam + LM + Lexicon	13.80	74.70	29.72	44.98	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	13.58	73.09	29.72	43.37	0.00
Mild Dysarthria Cohort					
1. Baseline Greedy CTC	49.54	135.71	7.56	91.18	36.97
2. CTC Prefix Beam Search	46.45	140.76	7.56	91.18	42.02
3. CTC + Blank Penalty ($\gamma = 0.2$)	49.22	139.50	7.56	91.18	40.76
4. CTC + Beam Search + LM	47.69	120.17	17.65	80.67	21.85
5. CTC + Beam + LM + Lexicon	47.55	119.33	17.65	80.25	21.43
6. Full Pipeline (Blank + Beam + LM + Lex)	46.91	121.43	15.97	82.77	22.69
Moderate Dysarthria Cohort					
1. Baseline Greedy CTC	60.94	108.13	36.18	63.82	8.13
2. CTC Prefix Beam Search	57.07	107.72	36.18	63.82	7.72
3. CTC + Blank Penalty ($\gamma = 0.2$)	59.95	108.94	35.37	64.63	8.94
4. CTC + Beam Search + LM	57.70	100.41	44.31	55.69	0.41
5. CTC + Beam + LM + Lexicon	57.70	100.41	44.31	55.69	0.41
6. Full Pipeline (Blank + Beam + LM + Lex)	56.53	101.63	43.90	56.10	1.63
Severe Dysarthria Cohort					
1. Baseline Greedy CTC	90.28	100.00	76.28	23.72	0.00
2. CTC Prefix Beam Search	76.53	100.00	62.45	37.55	0.00
3. CTC + Blank Penalty ($\gamma = 0.2$)	88.26	100.00	76.28	23.72	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	77.52	100.00	80.63	19.37	0.00

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